



LEADING PAYMENT TECHNOLOGY PROVIDER ACROSS ASEAN

raraglobal.net

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**Singapore-based payment tech provider
delivering innovative digital solutions for
banking and finance.**





Core Products

Innovative payment technology solutions powering digital transformation across ASEAN.



RARA eREMIT

Global Remittance Platform with single API integration and real-time cross-border transfers worldwide.



RARAPAY

Complete Digital Banking Suite including Cards, POS, Mobile Banking, and Risk & Fraud Management.



RARA TECH

Customized payment technology and outsourcing solutions for BFSI and retail sectors.



Our Services

Comprehensive consulting and investment solutions powering the future of digital finance across ASEAN.



RARA CONSULT

Expert guidance in mobile payments, e-commerce platforms, and blockchain innovation for financial institutions.



RARA INVEST

Strategic investments in promising fintech startups and established financial institutions across the region.



Full BFSI Solutions

End-to-end banking, financial services, and insurance solutions tailored to your business needs.



Global Clients Served

Trusted by clients across ASEAN, delivering reliable payment solutions to banks, financial institutions, and enterprises.



25+ Years of Experience

Over 25 years of domain expertise in payment technology, digital banking, and financial innovation across the region.



99.9% System Uptime

Enterprise-grade reliability with 99.9% uptime guarantee, ensuring uninterrupted payment processing for mission-critical operations.

Global Presence

▶ Singapore HQ

22 West Coast Road #03-01,
Varsity Park, Singapore 127446

▶ Nepal Office

Teendhara Road, Kamaladi-01,
Kathmandu, Nepal

▶ Regional Reach

Serving clients across ASEAN
with localized support and
expertise.

▶ 24/7 Support

Round-the-clock assistance
ensuring 99.9% system uptime
globally.





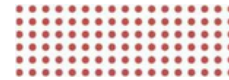
Nigrani - निगरानी

Agentic multi-agent ML system that scores every transaction in milliseconds and stops fraud before money moves.



Every transaction watched. Every fraud caught. In milliseconds.

The Problem Statement



Rule engines can't keep up with fraud

2.1%

block precision, only 2 of every 100 blocked transactions are real fraud

14%

false-positive rate, 50 innocent customers blocked per real fraud caught

NPR 34.6 B

of legitimate money frozen to catch just NPR 611M of fraud

219,753

innocent customers hit by false blocks

Fraud is a pattern problem, not a rule problem



Let's move to the proposed solution.

Solution Overview

Velocity Agent

spending bursts & speed

Geo Agent

impossible travel, VPN/datacenter IPs

Behaviour Agent

deviation from the customer's own profile

Device & Auth Agent

rooted devices, OTP, SIM-swap

Graph Agent

money-mule networks — finds entire rings

Synthesis Agent

fuses the 6 verdicts per transaction using per-type weights



Decision engine with thresholds

ALLOW / OTP-ONLY / BLOCK



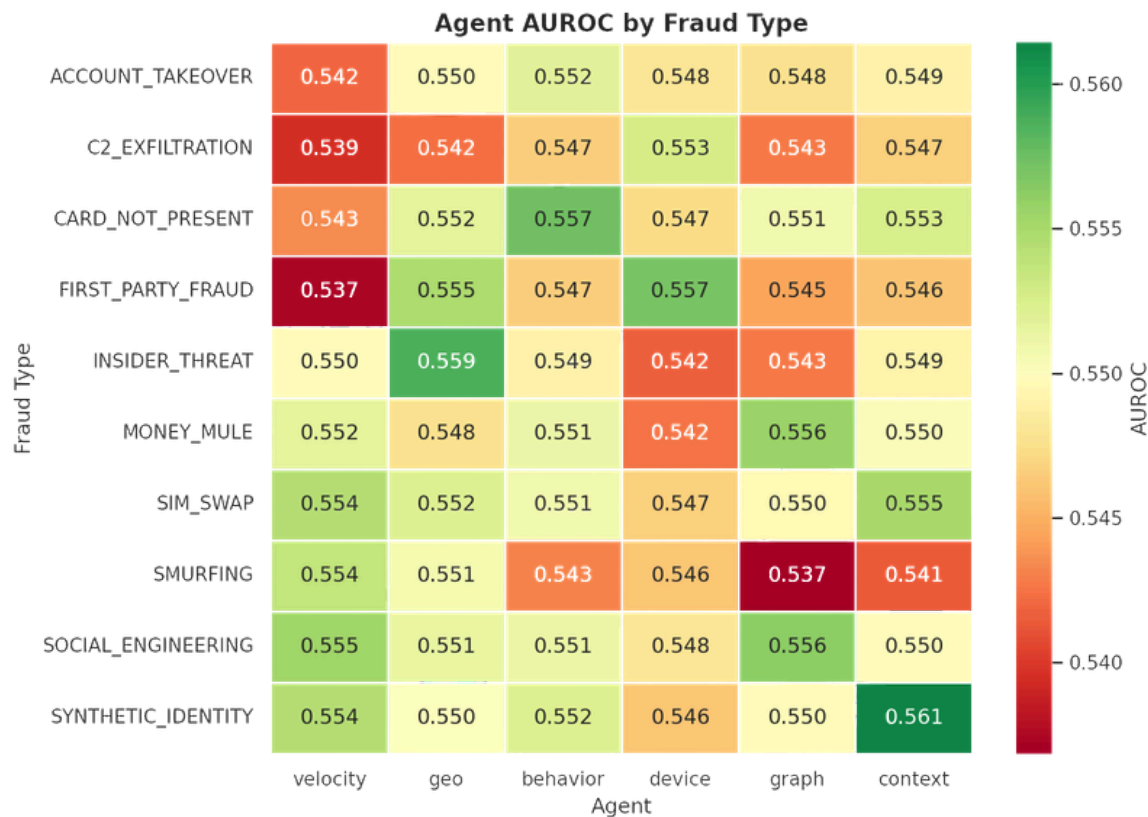
OTP Interlock

Dual-path SMS + email



SHAP Explainability

Top-5 reason codes on every single decision



Performance & Scalability

BLOCK PRECISION

72.7%

↗ 35x vs rule engine 2.1%

P95 LATENCY

3.71 ms

requirement < 800 ms · batch 0.0176 ms/txn

- Trained LightGBM boosters served in-process, no network hops, no model conversion
- Python FastAPI + production-grade Go rewrite (PostgreSQL 16, Redis, RabbitMQ) for high-throughput banking
- All 1.6M evaluation transactions scored in ~15 minutes on one device

Security



Secure Data Storage: PostgreSQL and Parquet for reliable, scalable data handling.



Strong Authentication: Secure password hashing with 8-hour session expiry.



Full Audit Trail: Logs every login and fraud decision for accountability.



Explainable Decisions: SHAP reason codes show why transactions were flagged.



Built-in Security: Strict validation, security headers, and protected error handling.

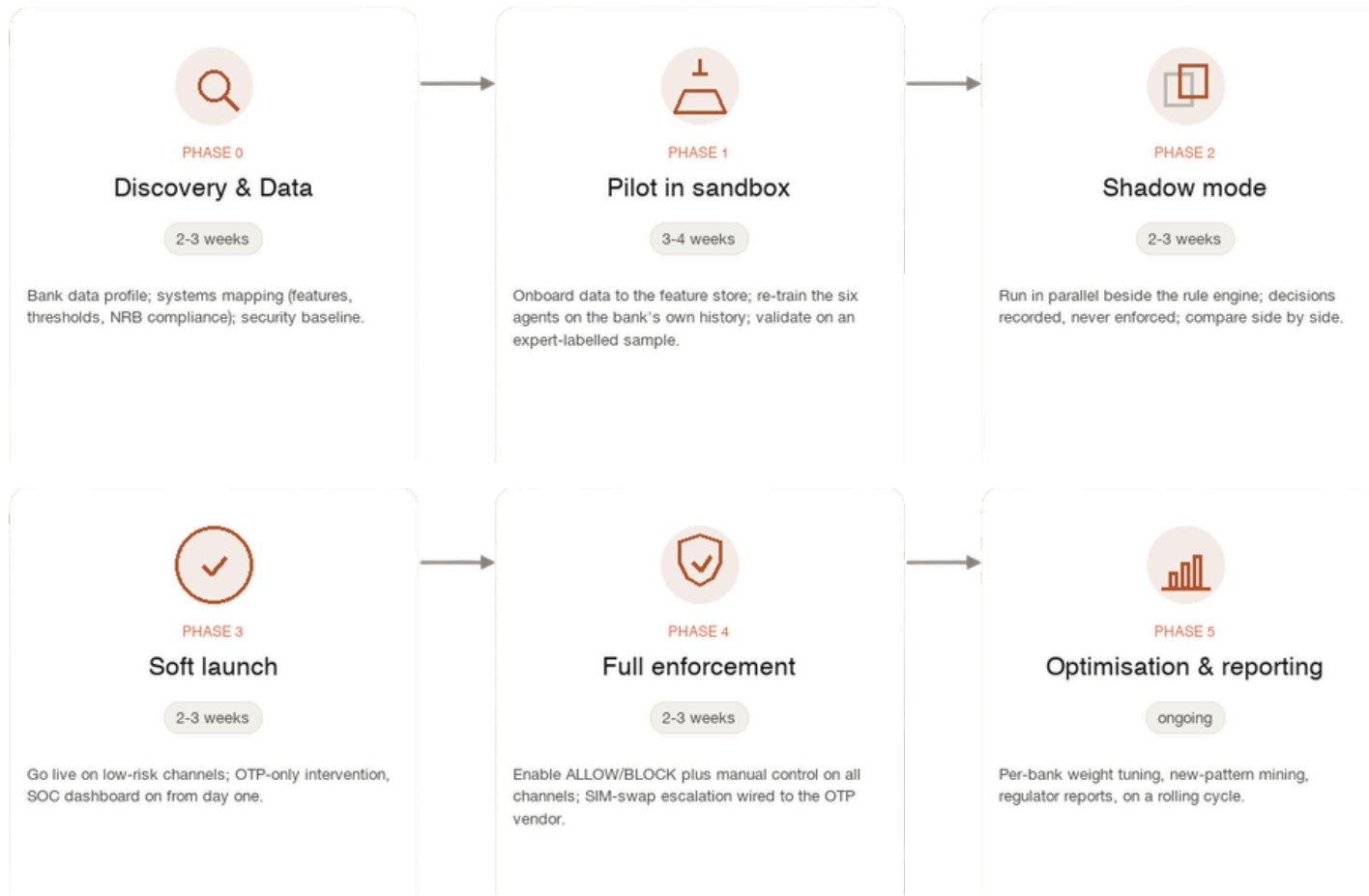


Business Advantages

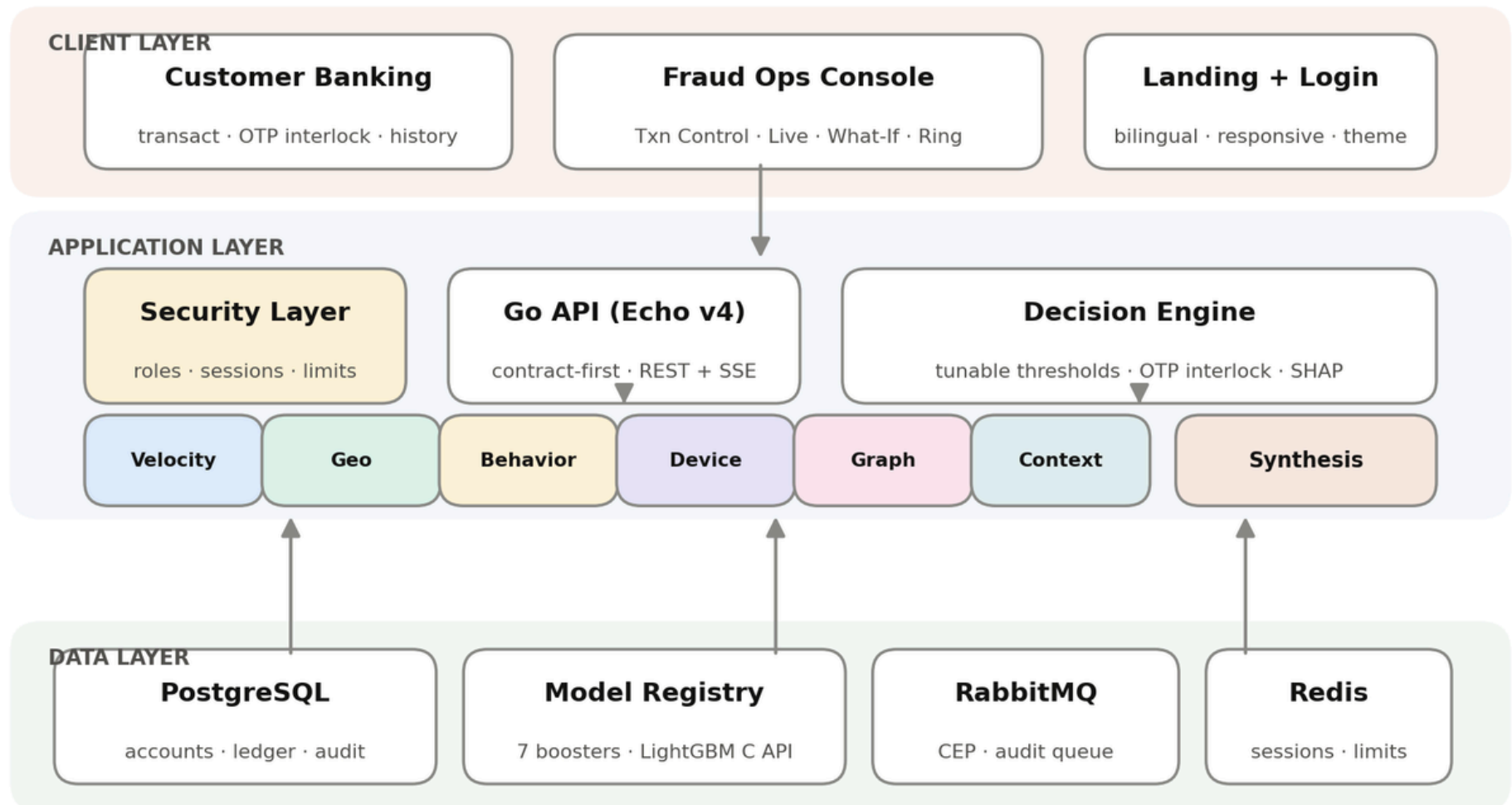
- ✓ 218x fewer false blocks
- ✓ NRB-compliant audit trail
- ✓ 74.7% vs 2.1% precision
- ✓ Mule-ring detection
- ✓ 0.06% false-positive rate
- ✓ Bilingual operations



Implementation Plan



System Overview





A security-hardened FastAPI service serves the trained agent fleet; the React console in front of it is what a bank's SOC team would actually operate.

Transaction verdict

Score: 95.8%

Agent verdicts

- Correct: 95.8%
- Incorrect: 4.2%

100 past agent verdicts

Agent	Verdict	Score
Agent 1	Correct	95.8%
Agent 2	Correct	95.8%
Agent 3	Correct	95.8%
Agent 4	Correct	95.8%
Agent 5	Correct	95.8%
Agent 6	Correct	95.8%
Agent 7	Correct	95.8%
Agent 8	Correct	95.8%
Agent 9	Correct	95.8%
Agent 10	Correct	95.8%

100 transactions with verdicts

Transaction	Verdict	Score
Trans 1	Correct	95.8%
Trans 2	Correct	95.8%
Trans 3	Correct	95.8%
Trans 4	Correct	95.8%
Trans 5	Correct	95.8%
Trans 6	Correct	95.8%
Trans 7	Correct	95.8%
Trans 8	Correct	95.8%
Trans 9	Correct	95.8%
Trans 10	Correct	95.8%

Why is this agent correct?

Score: 95.8%

100 transactions

Transaction	Verdict	Score
Trans 1	Correct	95.8%
Trans 2	Correct	95.8%
Trans 3	Correct	95.8%
Trans 4	Correct	95.8%
Trans 5	Correct	95.8%
Trans 6	Correct	95.8%
Trans 7	Correct	95.8%
Trans 8	Correct	95.8%
Trans 9	Correct	95.8%
Trans 10	Correct	95.8%

100 agents

Agent	Verdict	Score
Agent 1	Correct	95.8%
Agent 2	Correct	95.8%
Agent 3	Correct	95.8%
Agent 4	Correct	95.8%
Agent 5	Correct	95.8%
Agent 6	Correct	95.8%
Agent 7	Correct	95.8%
Agent 8	Correct	95.8%
Agent 9	Correct	95.8%
Agent 10	Correct	95.8%

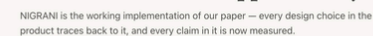
Live transaction monitor

Scored evaluation traffic streams in real time. Every row opens a full six-agent verdict – sub-scores, per-agent latency, and the final decision, checked against ground truth.

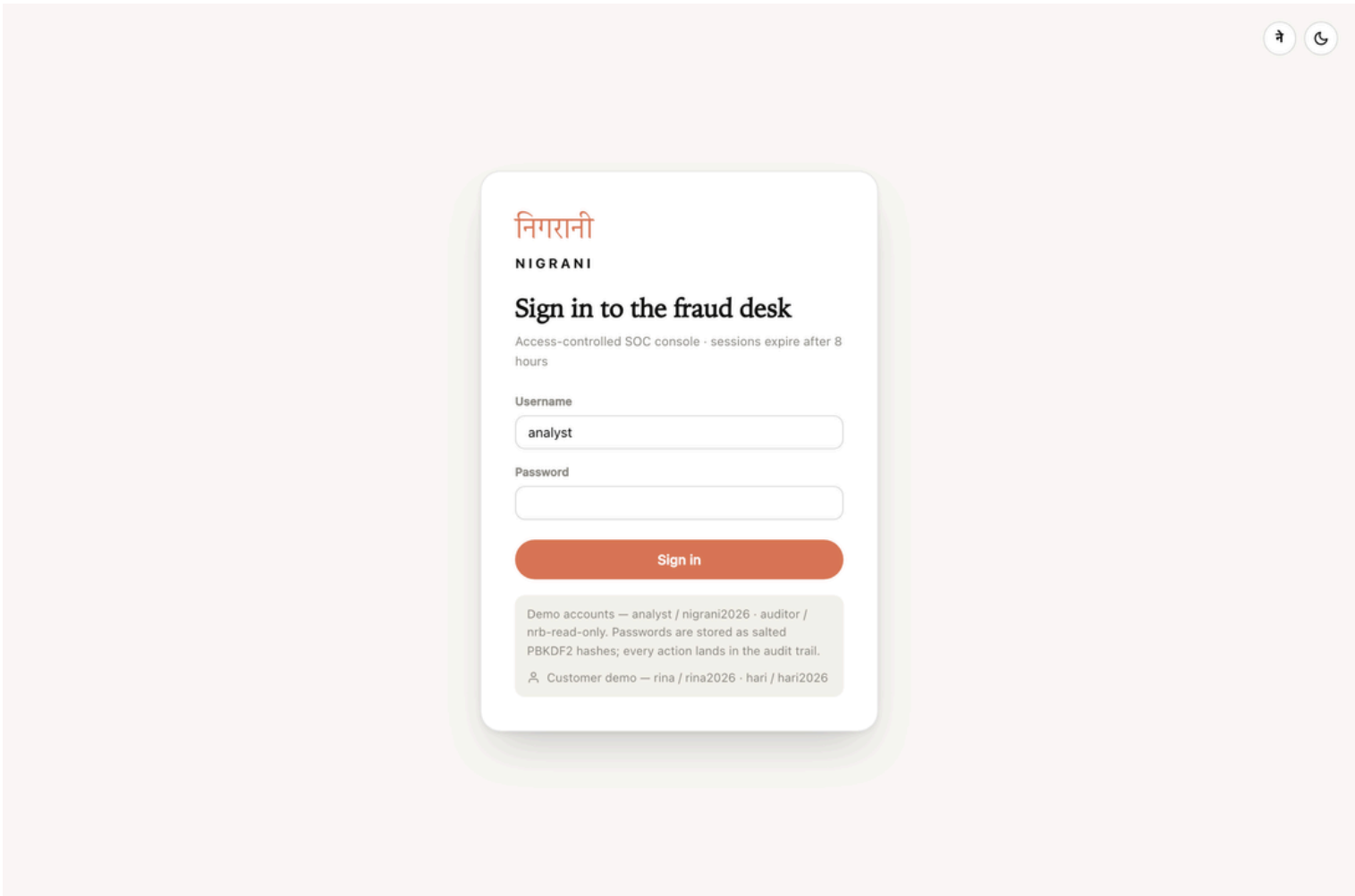
What-If laboratory

Take a legitimate transaction and inject an attack – impossible travel, a SIM swap, structuring at NPR 49,900 – then watch the responsible agent spike and the verdict flip to BLOCK. This is the live scoring API, not a mock.

Each agent sees only its own signal domain, so every verdict decomposes into who fired and why — the property that makes the system auditable.



Login Page

A mockup of the NIGRANI login page. The page has a light pink background. In the top right corner, there are two small circular icons: one with a 'ने' (Ne) symbol and another with a circular arrow. The main content is a white card with rounded corners and a subtle shadow. The card contains the NIGRANI logo (Hingraani in orange and NIGRANI in blue), the title "Sign in to the fraud desk", and a subtitle "Access-controlled SOC console - sessions expire after 8 hours". Below this are input fields for "Username" (containing "analyst") and "Password". A red "Sign in" button is positioned below the password field. At the bottom of the card, there is a grey box with demo account information: "Demo accounts — analyst / nigrani2026 - auditor / nrb-read-only. Passwords are stored as salted PBKDF2 hashes; every action lands in the audit trail." and "Customer demo — rina / rina2026 - hari / hari2026".

ने

निगरानी
NIGRANI

Sign in to the fraud desk

Access-controlled SOC console - sessions expire after 8 hours

Username

Password

Sign in

Demo accounts — analyst / nigrani2026 - auditor / nrb-read-only. Passwords are stored as salted PBKDF2 hashes; every action lands in the audit trail.

Customer demo — rina / rina2026 - hari / hari2026

नगरानीNAGRANI

Customer Banking

rina · ACC-8712448

Sign out

Welcome, rina

Demo account · Chitwan · LOW risk tier

Available balance
NPR 53,63,100

NEW TRANSACTION

Amount (NPR)

10000

Payment channel

ESEWA_P2P

Recipient

e.g. MERCH-1024 or ACC-...

First time sending to this recipient

Transaction context (simulated for the demo)

Transacting from

Home city

On VPN / proxy network

Routed / jailbroken device

Late night (01:00-04:00)

SIM recently changed

Fraud checks to run

All checks run by default. Turn one off to send this transaction without it — handy for testing the others on their own while a check is being fixed.

Velocity

Geo

Behavior

Device & Auth

Graph

Context (S_a)

Send transaction

RECENT TRANSACTIONS

REF	AMOUNT (NPR)	PAYMENT CHANNEL	STATUS
FF3B34	NPR 4,600	MOBILE_TOPUP	Awaiting OTP
41D3C4	NPR 36,000	IPS_TRANSFER	Blocked
90A1B0	NPR 12,500	MOBILE_WALLET	Blocked
7E072D	NPR 4,200	ESEWA_P2P	Blocked
C9A923	NPR 4,600	MOBILE_TOPUP	Blocked
F74287	NPR 4,600	MOBILE_TOPUP	Awaiting OTP
C19594	NPR 36,000	IPS_TRANSFER	Blocked
8B8D2D	NPR 12,500	MOBILE_WALLET	Blocked
D74318	NPR 4,200	ESEWA_P2P	Blocked
9D270E	NPR 4,600	MOBILE_TOPUP	Blocked
0E036D	NPR 4,600	MOBILE_TOPUP	Awaiting OTP
25A001	NPR 36,000	IPS_TRANSFER	Blocked
71B202	NPR 12,500	MOBILE_WALLET	Blocked
995408	NPR 4,200	ESEWA_P2P	Blocked
7FC76A	NPR 4,600	MOBILE_TOPUP	Awaiting OTP
A98573	NPR 36,000	IPS_TRANSFER	Blocked
4EC95A	NPR 12,500	MOBILE_WALLET	Blocked
1414CA	NPR 4,200	ESEWA_P2P	Completed
0B7C32	NPR 4,600	MOBILE_TOPUP	Awaiting OTP
04F9C0	NPR 3,800	MOBILE_TOPUP	Awaiting OTP

Customer OTP Feature


NIGRANI
Customer Banking


rina · ACC-8712448





Sign out


Welcome, rina
Demo account · Chitwan · LOW risk tier

Available balance
NPR 53,58,500

NEW TRANSACTION

Amount (NPR)

Payment channel
MOBILE_TOPUP

Recipient


First time sending to this recipient

Transaction context (simulated for the demo)
Transacting from


On VPN / proxy network


Routed / jailbroken device


Late night (01:00–04:00)


SIM recently changed

Fraud checks to run
All checks run by default. Turn one off to send this transaction without it — handy for testing the others on their own while a check is being fixed.

Velocity ☒
Geo ☒
Behavior ☒
Device & Auth ☒
Graph ☒

Context (S_a) ☒


Send transaction

Awaiting OTP
Additional verification required

Why


otp_attempts_max = 2


Amount is 0.0x the account's 30-day average


total_amount_1h_npr = 1128000.0

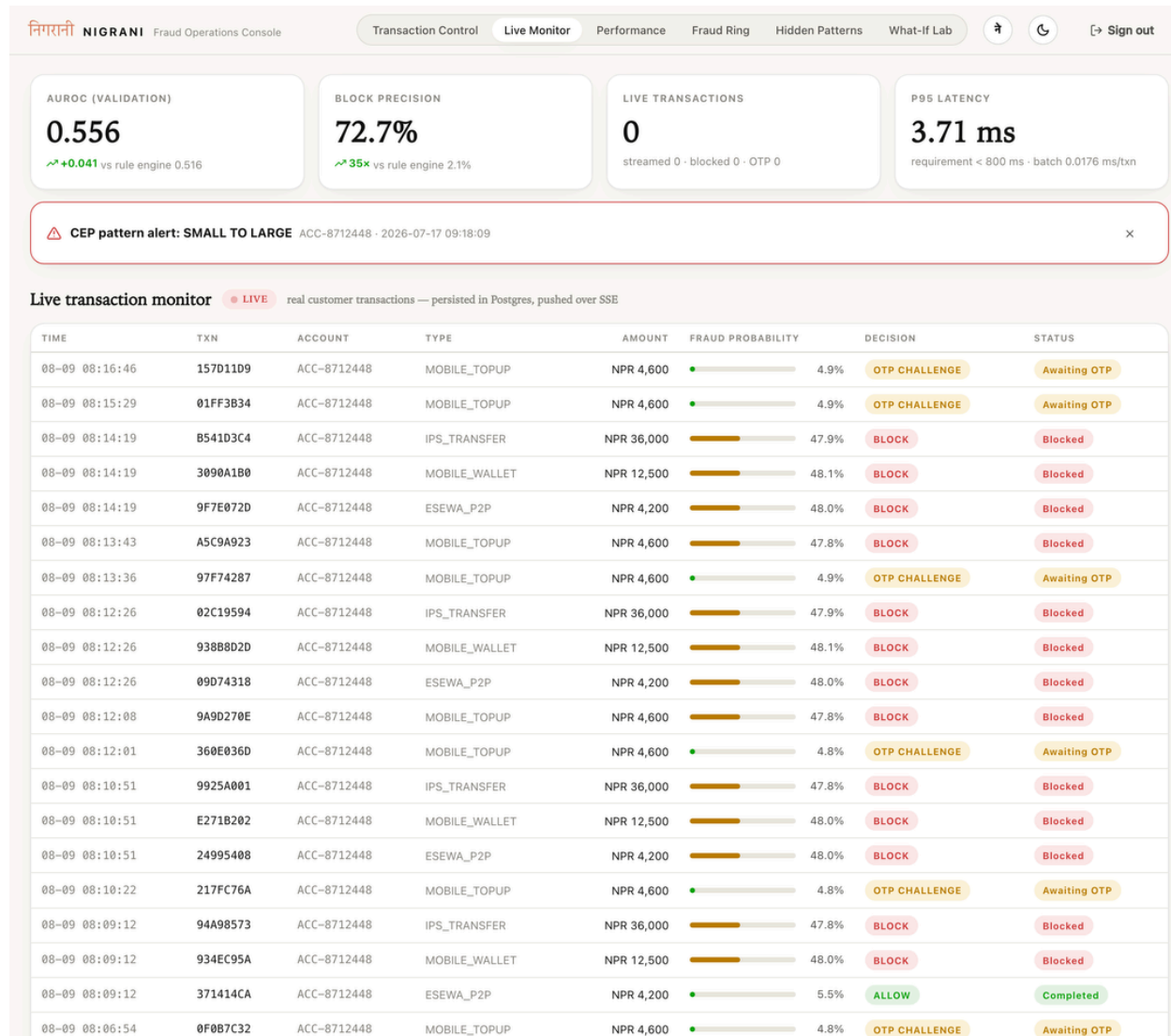
TRANS	AMOUNT (NPR)	PAYMENT CHANNEL	STATUS
		MOBILE_TOPUP	Awaiting OTP
		MOBILE_TOPUP	Awaiting OTP
		IPS_TRANSFER	Blocked
90A1B0	NPR 12,500	MOBILE_WALLET	Blocked
7E072D	NPR 4,200	ESEWA_P2P	Blocked
C9A923	NPR 4,600	MOBILE_TOPUP	Blocked
F74287	NPR 4,600	MOBILE_TOPUP	Awaiting OTP
C19594	NPR 36,000	IPS_TRANSFER	Blocked
8B8D2D	NPR 12,500	MOBILE_WALLET	Blocked
D74318	NPR 4,200	ESEWA_P2P	Blocked
9D270E	NPR 4,600	MOBILE_TOPUP	Blocked


Verify it's you
A one-time code was sent via SMS + EMAIL.
Demo code: 123456

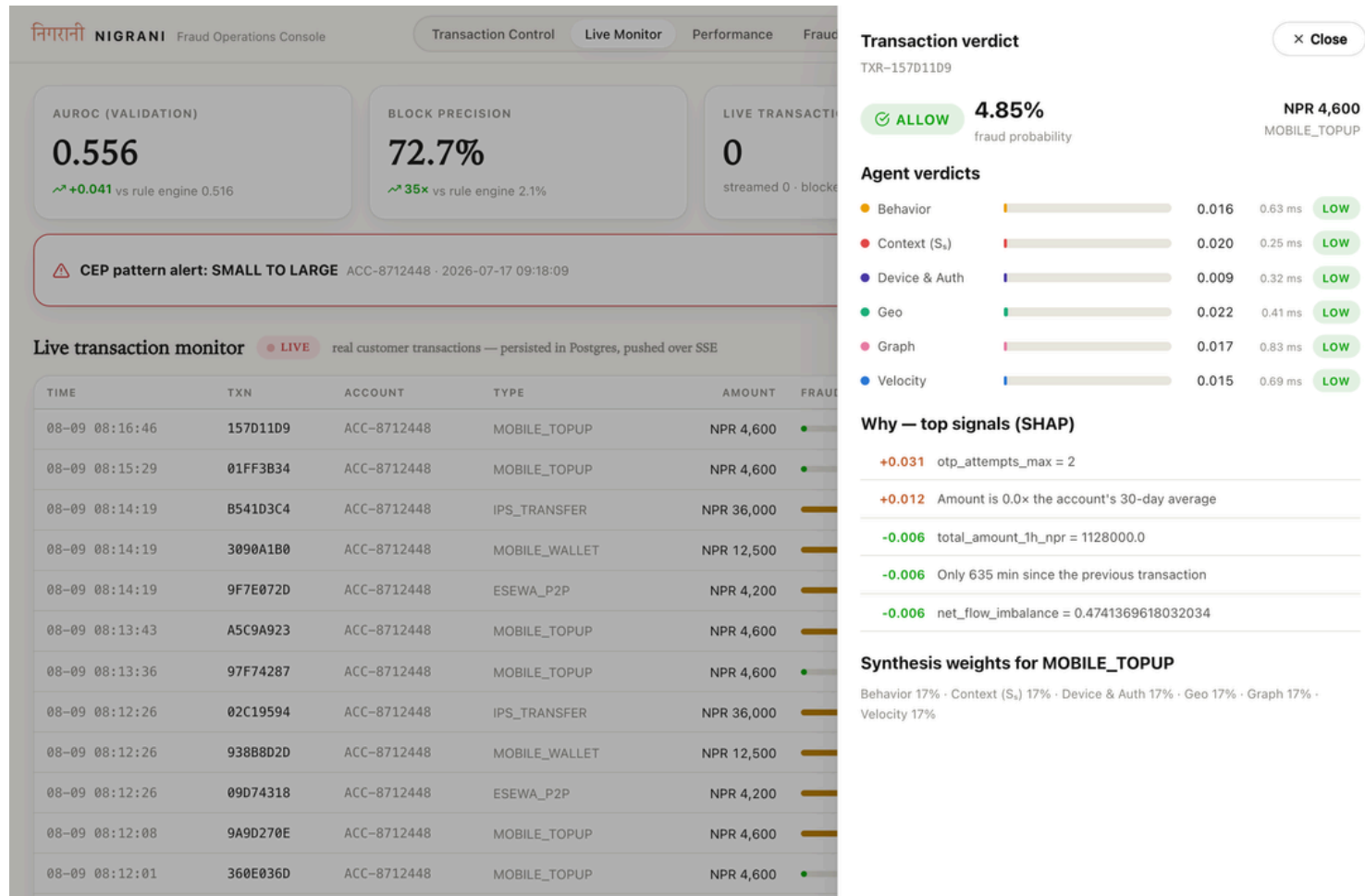

Verify

Cancel

Admin Live Monitoring



Admin Transaction Verdict



Admin Transaction Control

निराणी NIGRANI Fraud Operations Console
Transaction Control
Live Monitor
Performance
Fraud Ring
Hidden Patterns
What-If Lab
Sign out

LIVE CUSTOMER TRANSACTIONS
297
297 transactions

30 AWAITING OTP
30
auto-refreshing

DECISION THRESHOLDS

Tune where the fused fraud probability turns into an OTP challenge or a hard block. Applies instantly to every new verdict (customer plane, What-If Lab, live scoring), is persisted, and every change lands in the NRB audit trail.

OTP challenge above
6.05%
Block above
8.65%

Apply
Restore model defaults

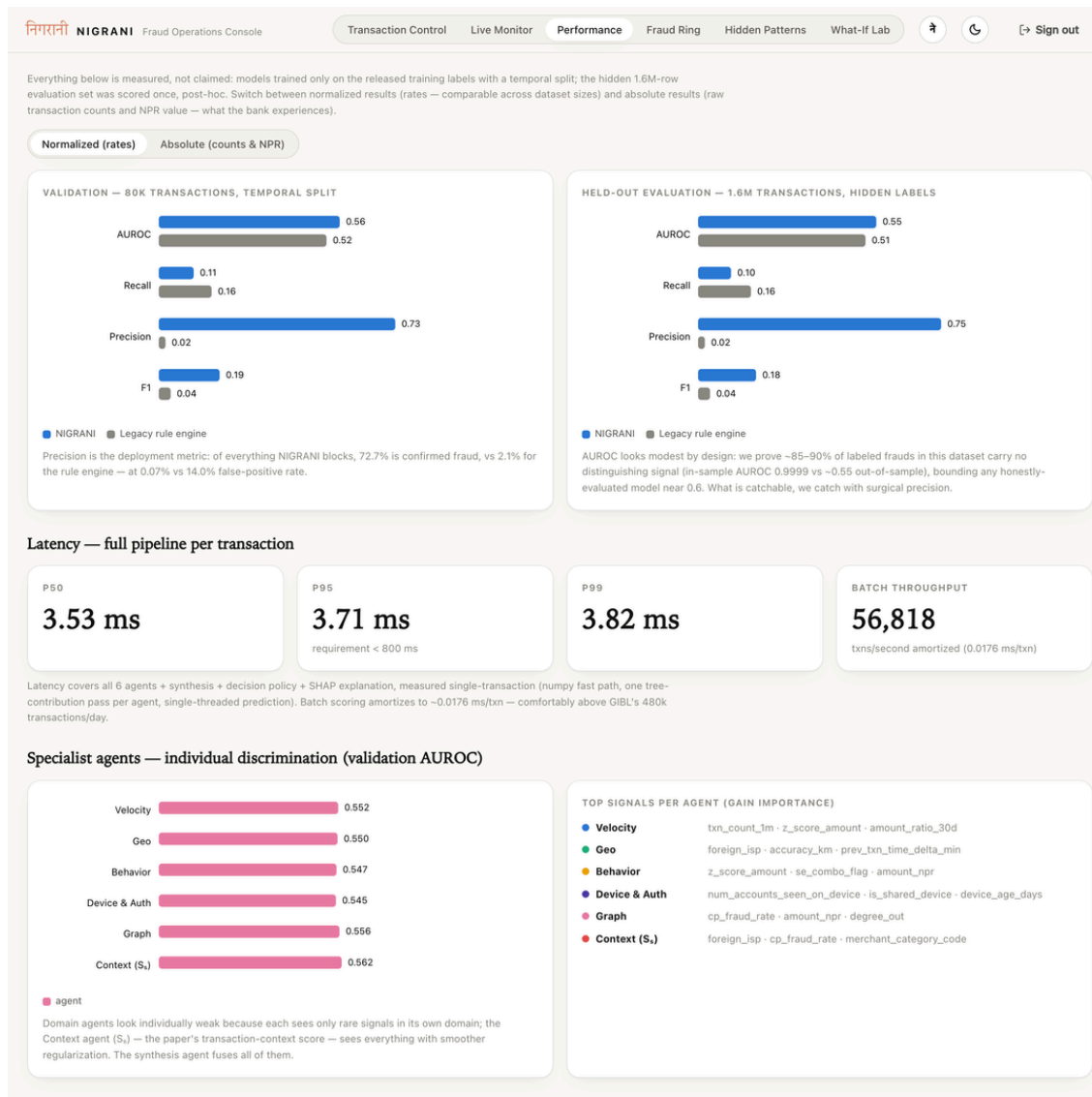
score $\geq$ 6.05% $\rightarrow$ OTP challenge - score $\geq$ 8.65% $\rightarrow$ block - model defaults: 6.05% / 8.65%

Live customer transactions

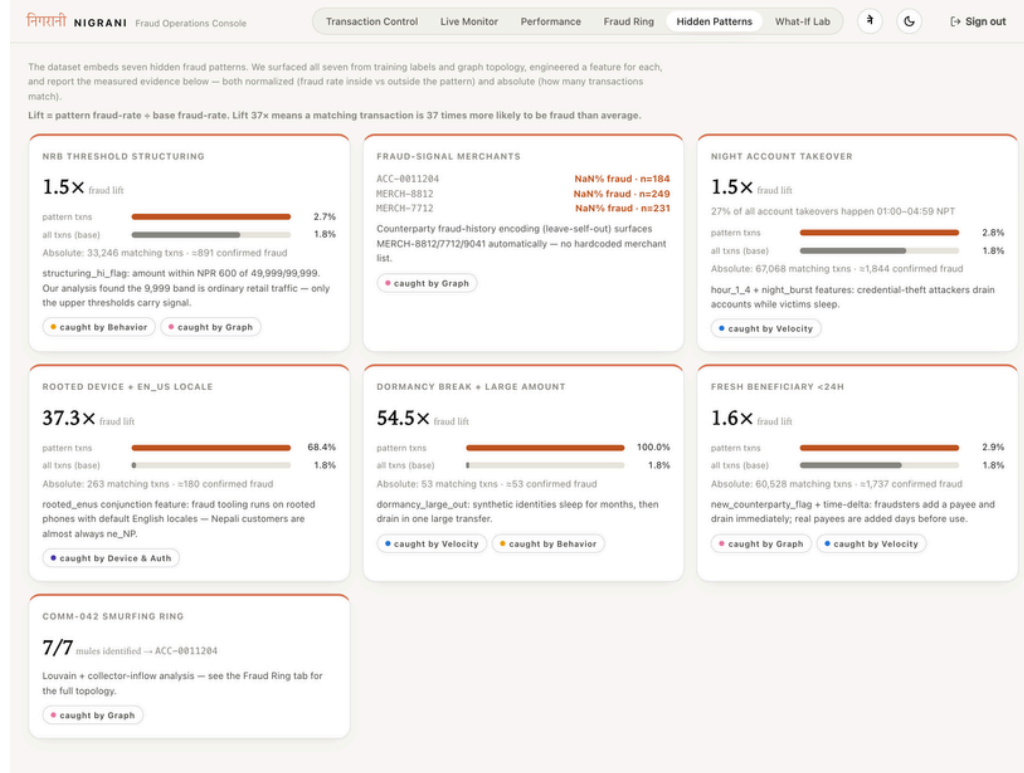
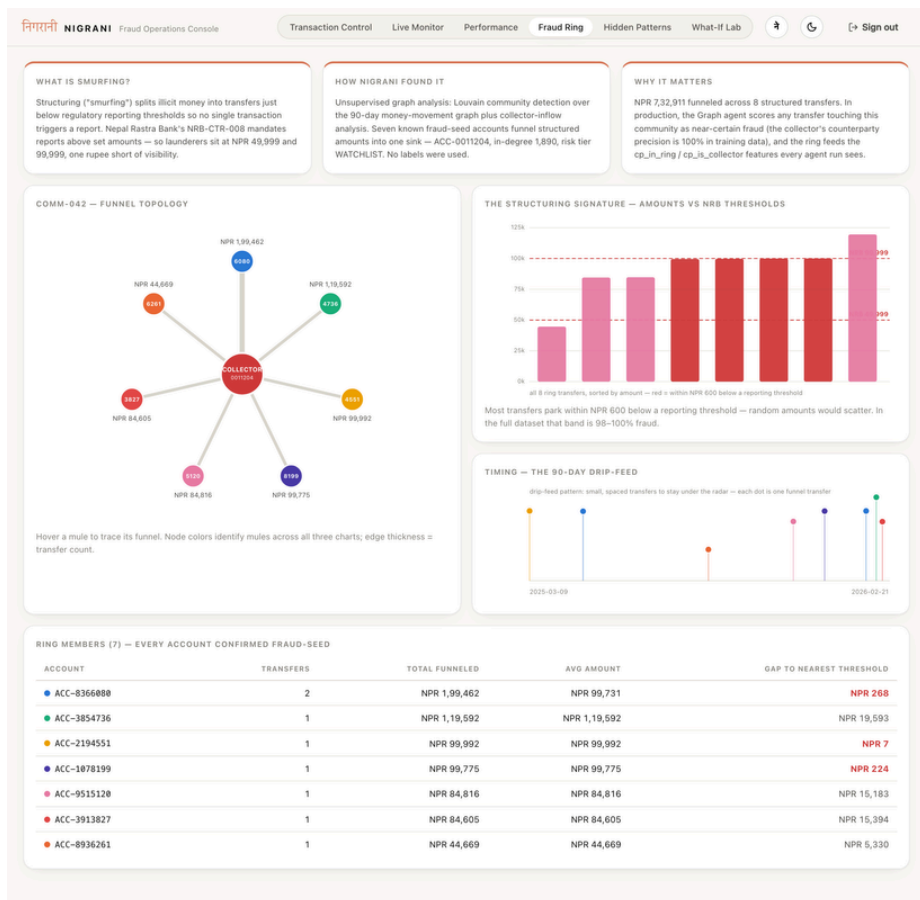
Real-time feed of customer-initiated transactions. A SOC analyst can manually halt an awaiting-OTP transaction, or freeze one that has already completed — pending investigation.

TIME	CUSTOMER	REF	AMOUNT	CHANNEL	DECISION	STATUS	ACTION
08:16:46	rina	7D11D9	NPR 4,600	MOBILE_TOPUP	OTP CHALLENGE	Awaiting OTP	Halt
08:15:29	rina	FF3B34	NPR 4,600	MOBILE_TOPUP	OTP CHALLENGE	Awaiting OTP	Halt
08:14:19	rina	41D3C4	NPR 36,000	IPS_TRANSFER	BLOCK	Blocked	—
08:14:19	rina	90A1B0	NPR 12,500	MOBILE_WALLET	BLOCK	Blocked	—
08:14:19	rina	7E072D	NPR 4,200	ESEWA_P2P	BLOCK	Blocked	—
08:13:43	rina	C9A923	NPR 4,600	MOBILE_TOPUP	BLOCK	Blocked	—
08:13:36	rina	F74287	NPR 4,600	MOBILE_TOPUP	OTP CHALLENGE	Awaiting OTP	Halt
08:12:26	rina	C19594	NPR 36,000	IPS_TRANSFER	BLOCK	Blocked	—
08:12:26	rina	8B8D2D	NPR 12,500	MOBILE_WALLET	BLOCK	Blocked	—
08:12:26	rina	D74318	NPR 4,200	ESEWA_P2P	BLOCK	Blocked	—
08:12:08	rina	9D270E	NPR 4,600	MOBILE_TOPUP	BLOCK	Blocked	—
08:12:01	rina	0E036D	NPR 4,600	MOBILE_TOPUP	OTP CHALLENGE	Awaiting OTP	Halt
08:10:51	rina	25A001	NPR 36,000	IPS_TRANSFER	BLOCK	Blocked	—
08:10:51	rina	71B202	NPR 12,500	MOBILE_WALLET	BLOCK	Blocked	—

Admin Performance Page



Admin Fraud Ring and Hidden Patterns



Admin What-if Lab

निरगानी NIGRANI Fraud Operations Console
Transaction Control Live Monitor Performance Fraud Ring Hidden Patterns What-if Lab
Sign out

The What-If laboratory is the live scoring API, not a mock: pick a legitimate transaction, inject an attack signature, and the five domain agents + context agent re-score it in milliseconds. Derived features (z-scores, structuring distances, travel speed) are recomputed server-side so every scenario stays physically coherent. The signed numbers on each reason are SHAP values — each feature's contribution to this exact verdict.

SCENARIO BUILDER

Base transaction (a real ALLOW from the evaluation stream)

TXN-20250409-DAFE905D

Attack scenarios — one click sets the full signal bundle

Impossible travel + VPN
Night burst attack

Structuring (NPR 49,900)
SIM-swap + compromised device

Dormant account drain

The amount is parked NPR 99 below the 49,999 NRB reporting threshold, going to a first-time recipient. Deliberately sub-threshold transfers are how launderers stay invisible to NRB-CTR-008 reporting — the COMM-042 ring uses exactly this trick.

Expected: Behavior agent fires · typology SMURFING

Amount (NPR)

49900

VPN detected

(keep)

Impossible travel

(keep)

Night transaction

(keep)

Txns in last minute

(keep)

New recipient

Yes

Rooted device

(keep)

Dormancy break

(keep)

Score transaction
Reset

VERDICT — BEFORE VS AFTER

before ALLOW 6.03%

after BLOCK 50.46%

the injected signals moved this transaction from ALLOW to BLOCK (+44.4% probability) in 4.83 ms

Behavior	0.989	0.57 ms	CRITICAL
Context (S ₂)	0.019	0.2 ms	LOW
Device & Auth	0.015	0.31 ms	LOW
Geo	0.020	0.39 ms	LOW
Graph	0.031	0.8 ms	LOW
Velocity	0.121	0.57 ms	LOW

TOP SIGNALS — SHAP CONTRIBUTIONS TO THIS VERDICT

- 0.648 Amount is 18.3 standard deviations above this account's 30-day norm
- 0.036 Amount within NPR 600 of the 49,999/99,999 NRB thresholds (98-100% fraud precision)
- 0.017 amount_npr = 49900.0
- 0.017 Recipient has an elevated historical fraud rate
- +0.014 Amount is 11.7× the account's 30-day average

Predicted typology: **MONEY MULE** (expected: SMURFING)

Let's Connect



Our Promise

Empowering Digital Payments, Transforming Financial Futures across ASEAN and beyond.



Get Started Today

Ready to transform your payment infrastructure? Contact RARA Global to discover how our innovative solutions can elevate your business.

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